Spandan Ghoshal

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Research Interests

High-Dimensional Statistics · **Change-Point Analysis** · **Nonparametric Methods** · **Two-Sample Testing** · **Random Graphs and Network Data** · **Probabilistic Combinatorics**

Education

Indian Statistical Institute — **Kolkata**
Ph.D. in Statistics (Senior Research Fellow) — *2023–Present*
Supervisor: Prof. Anil K. Ghosh. Research on high-dimensional change-point detection and nonparametric testing.

Indian Statistical Institute — **Kolkata**
Master of Statistics (M.Stat), ***78.9% (First Division with Distinction)*** — *2023*
Specialization: Computational Statistics.

Presidency University — **Kolkata**
B.Sc. in Statistics, ***CGPA: 8.41/10, Ranked 1st in Department*** — *2021*
Batch 2018–2021.

Howrah Zilla School — **Howrah**
Class XII (WBCHSE), ***94.0%*** — *2018*

Howrah Vivekananda Institution — **Howrah**
Class X (WBBSE), ***88.4%*** — *2016*

Academic Achievements

- Qualified **UGC-NET** (Statistics), June 2026, at the **99.33 percentile**; eligible for Assistant Professor.
- Secured **All India Rank 1** in ISI JRF Entrance Examination (2023).
- Secured **All India Rank 9** in IIT-JAM Statistics (2021).
- Secured **All India Rank 10** in ISI M.Stat Entrance Examination (2021).
- Recipient of the ISI Fellowship for Ph.D. and M.Stat programs.

Publications & Research Work

Manuscripts

Under Revision: **Ghoshal, S.**, Banerjee, B., and Ghosh, A. K. "On High-Dimensional Change-Point Detection Based on Pairwise Distances." *Statistica Sinica* (major revision). arXiv:2511.10078.

Preprint: Athreya, S., Bhattacharya, B. B., Dahiya, Y., Das, K., **Ghoshal, S.**, Maniyar, S., and Roy, E. S. "Third and Fourth Moment Phenomena for Monochromatic Edges and Applications to Graph-Based Two-Sample Tests." arXiv:2610.29215, 2026. *Initiated at the ICTS-Networks Workshop.*

In Preparation: **Ghoshal, S.** and Ghosh, A. K. "Universal consistency of the distribution-free multivariate runs test based on the shortest Hamiltonian path."

Teaching Experience

Statistical Learning I — **BSDS 3rd Year, ISI Kolkata**
Teaching Assistant — *Jul–Dec 2026*
Instructor: Prof. Anil K. Ghosh.

Nonparametric and Sequential Methods — **B.Stat 3rd Year, ISI Kolkata**
Teaching Assistant — *Jan–Jun 2026*
Instructor: Prof. Anil K. Ghosh.

Statistical Structure in Data — **PGDBA 1st Year, ISI Kolkata**
Teaching Assistant — *Jul–Dec 2025*
Instructor: Prof. Anil K. Ghosh.

Statistical Data Analysis — **ISEC, ISI Kolkata**
Co-Instructor — *Jan–Jun 2025*
Co-taught with Prof. Sumitra Purakayastha.

Descriptive Statistics I — **ISEC, ISI Kolkata**
Co-Instructor — *Jul–Dec 2024*
Co-taught with Prof. Partha Sarathi Mukherjee.

NCM Workshop on Statistical Learning — **ISI Kolkata**
Teaching Assistant — *Jul 2024*

Vectors & Matrices II — **B.Stat 1st Year, ISI Kolkata**
Co-Instructor — *Jan–Jun 2024*
Co-taught with Prof. Sumitra Purakayastha.

Conferences & Workshops

Modern Trends in Probability and Statistics (Basu–Bahadur Memorial Conference) — **ISI Kolkata**
Speaker — *Dec 2026*
December 18–22, 2026 (upcoming).

Program on Selective Inference — **Indian School of Business, Hyderabad**
Selected Participant — *Nov 2026*
November 2–6, 2026 (upcoming).

European Meeting of Statisticians 2026 — **Lugano, Switzerland**
Contributed Speaker — *Aug 2026*
Talk: "High-Dimensional Change-Point Detection Based on Pairwise Distances" (Paper ID: 74).

ICTS-Networks Workshop: "Challenges in Networks" — **ICTS-TIFR**
Speaker & Research Group Member — *Apr 2026*
Working group (Week 1) and talk (Week 2) on joint research with Prof. Athreya and Prof. Bhattacharya.

Calcutta Triennial Symposium 2024 — **Kolkata, India**
Participant — *Dec 2024*

Academic Service

Journal Referee: Served as a referee for *Sankhya B: The Indian Journal of Statistics* and the *Indian Journal of Pure and Applied Mathematics*.

Technical Skills

Languages: R, Python, C, LaTeX
Tools: Overleaf, RStudio, MS Office

Extra-Curricular Activities

Music: Indian Classical Flute (under Sri Pankaj Banerjee) and Harmonica (under Sri Sukanta Ghoshal).